

Exact posterior

$$\pi(\boldsymbol{\vartheta} \mid \mathbf{y}) \propto \pi(\boldsymbol{\vartheta}) \pi(\mathbf{y} \mid \boldsymbol{\vartheta})$$

Laplace's method

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Joint Gaussian approximation

(a) Mode & curvature
 $N_m(\boldsymbol{\vartheta}^*, \boldsymbol{\Omega})$

(b) VB mean correction
 $N_m(\tilde{\boldsymbol{\vartheta}}, \boldsymbol{\Omega})$

Model comparison

log-evidence,
Bayes factors, LOO-CV